MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER PRACTICAL EXAMINATION
(2024 ADMISSION ONWARDS)

MG3CCRBCA201- Database Management Systems

Duration of the Examination: **1.5 Hrs** Maximum Marks: **35**

List of Questions

***** All questions belong to the **Apply** level and are aligned with Course Outcome 5 (**CO5**).

1. *Database Schema*
   Supplier(SID, SName, Branch, City)
   Part(PID, PName, Color, Cost)
   Supplies(SID, PID, Quantity)

   Write SQL queries for the following:
   a) Find the minimum, maximum, average, and total cost of all parts.
   b) Display each color with total value (Cost × Quantity) of supplied parts using GROUP BY.
   c) List only those colors where total value exceeds 5000 using HAVING.

2. *Database Schema*
   Supplier(SID, SName, Branch, City)
   Part(PID, PName, Color, Cost)
   Supplies(SID, PID, Quantity)
   a) Count the total number of parts.
   b) Display each supplier with the total quantity of parts they supplied.
   c) Show suppliers who supplied more than 50 units (using HAVING).
3. *Database Schema*
   Supplier(SID, SName, Branch, City)
   Part(PID, PName, Color, Cost)
   Supplies(SID, PID, Quantity)
   a) Display each branch with total number of suppliers (GROUP BY).
   b) Show only branches with more than 2 suppliers (HAVING).
   c) Sort results by branch name in ascending order.
4. *Database Schema*
   Supplier(SID, SName, Branch, City)
   Part(PID, PName, Color, Cost)
   Supplies(SID, PID, Quantity)

a) List all parts sorted by `PName` (A–Z).
b) For parts with same name, sort by `Cost` in descending order.
c) Find the average cost of all Black-colored parts.

5. *Database Schema*
Supplier(SID, SName, Branch, City)
Part(PID, PName, Color, Cost)
Supplies(SID, PID, Quantity)
a) Retrieve average cost of parts supplied by 'Mike'.
b) Display each supplier with count of distinct parts supplied.
c) Show only suppliers supplying more than 3 different parts.
6. *Database Schema*
Supplier(SID, SName, Branch, City)
Part(PID, PName, Color, Cost)
Supplies(SID, PID, Quantity)
a) Display each city with total suppliers.
b) Show only cities with more than 2 suppliers.
c) Sort results by number of suppliers (highest to lowest).
7. *Database Schema*
Supplier(SID, SName, Branch, City)
Part(PID, PName, Color, Cost)
Supplies(SID, PID, Quantity)
a) Display each part with total revenue generated (Cost × Quantity).
b) Show only those parts generating revenue above 10,000.
c) Sort by revenue in descending order.
8. *Database Schema*
Supplier(SID, SName, Branch, City)
Part(PID, PName, Color, Cost)
Supplies(SID, PID, Quantity)
a) Display each color with number of parts in that color.
b) Display each branch with number of suppliers.
c) Show colors with more than 2 parts and branches with more than 2 suppliers.
9. *Database Schema*
Supplier(SID, SName, Branch, City)
Part(PID, PName, Color, Cost)
Supplies(SID, PID, Quantity)
a) Find min, max, and avg cost of parts supplied by each supplier.
b) Display only suppliers whose avg part cost is above 2000.
c) Sort these suppliers by average cost.
10. *Database Schema*
Supplier(SID, SName, Branch, City)
Part(PID, PName, Color, Cost)
Supplies(SID, PID, Quantity)
a) Display each supplier with total quantity and total cost supplied.
b) Show only suppliers contributing quantity > 100 units.
c) Sort results by total quantity (DESC) and supplier name (ASC)